

Diag

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 40 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If $\sin \theta = \frac{1}{2}$, then the value of $\cos \theta$ is

- (A) (A) 1
- (B) (B) 0
- (C) (C) $\frac{\sqrt{3}}{2}$
- (D) (D) $\frac{1}{\sqrt{2}}$

2. The value of $\tan 45^\circ$ is

- (A) (A) 0
- (B) (B) 1
- (C) (C) $\sqrt{3}$
- (D) (D) $\frac{1}{\sqrt{3}}$

3. If $\cos A = \frac{4}{5}$, then $\tan A$ is

- (A) (A) $\frac{3}{4}$
- (B) (B) $\frac{5}{3}$
- (C) (C) $\frac{4}{3}$
- (D) (D) $\frac{3}{5}$

4. What is the value of $\sin^2 30^\circ + \cos^2 30^\circ$?

- (A) (A) 0
- (B) (B) $1/2$
- (C) (C) 1
- (D) (D) 2

5. If $3 \cot \theta = 2$, then the value of $\tan \theta$ is

- (A) (A) $2/3$

- (B) (B) $1/3$
- (C) (C) 3
- (D) (D) $3/2$

6. The value of $\frac{\sin 18^\circ}{\cos 72^\circ}$ is

- (A) (A) 0
- (B) (B) 1
- (C) (C) 2
- (D) (D) -1

7. If $\sec \theta = 2$, then θ is

- (A) (A) 30°
- (B) (B) 45°
- (C) (C) 60°
- (D) (D) 90°

8. The value of $(1 + \tan^2 \theta) \cos^2 \theta$ is

- (A) (A) 1
- (B) (B) 0
- (C) (C) 2
- (D) (D) -1

9. If $\sin A = \cos A$, then the value of A is

- (A) (A) 30°
- (B) (B) 45°
- (C) (C) 60°
- (D) (D) 0°

10. The value of $\operatorname{cosec}^2 \theta - 1$ is equal to

- (A) (A) $\tan^2 \theta$
- (B) (B) $\sin^2 \theta$
- (C) (C) $\cos^2 \theta$
- (D) (D) $\cot^2 \theta$

11. If $\sin \theta = \frac{1}{2}$, then the value of $\cos \theta$ is

- (A) (A) 1
- (B) (B) 0
- (C) (C) $\frac{\sqrt{3}}{2}$

(D) $(D) \frac{1}{\sqrt{2}}$

12. The value of $\tan 45^\circ$ is

(A) (A) 0

(B) (B) 1

(C) (C) $\sqrt{3}$

(D) (D) $\frac{1}{\sqrt{3}}$

13. If $\cos A = \frac{4}{5}$, then $\tan A$ is

(A) (A) $\frac{3}{4}$

(B) (B) $\frac{5}{3}$

(C) (C) $\frac{4}{3}$

(D) (D) $\frac{3}{5}$

14. What is the value of $\sin^2 30^\circ + \cos^2 30^\circ$?

(A) (A) 0

(B) (B) $1/2$

(C) (C) 1

(D) (D) 2

15. If $3 \cot \theta = 2$, then the value of $\tan \theta$ is

(A) (A) $2/3$

(B) (B) $1/3$

(C) (C) 3

(D) (D) $3/2$

16. The value of $\frac{\sin 18^\circ}{\cos 72^\circ}$ is

(A) (A) 0

(B) (B) 1

(C) (C) 2

(D) (D) -1

17. If $\sec \theta = 2$, then θ is

(A) (A) 30°

(B) (B) 45°

(C) (C) 60°

(D) (D) 90°

18. Assertion (A): For an angle θ , $\sin^2 \theta + \cos^2 \theta = 1$. Reason (R): This identity holds true for all values of θ where $0^\circ \leq \theta \leq 90^\circ$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
19. Assertion (A): The value of $\tan 45^\circ$ is 1. Reason (R): $\tan \theta = \frac{\sin \theta}{\cos \theta}$ and $\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
20. Assertion (A): The value of $\sin \theta$ can be 1.5 for some angle θ . Reason (R): The range of $\sin \theta$ is $[-1, 1]$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
21. Assertion (A): $\sec^2 \theta - \tan^2 \theta = 1$. Reason (R): $1 + \tan^2 \theta = \sec^2 \theta$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
22. Assertion (A): If $\cos A = \frac{1}{2}$, then $\sin A = \frac{\sqrt{3}}{2}$. Reason (R): $\sin^2 A + \cos^2 A = 1$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

1. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .
2. Show that $\tan A \cdot \tan(90^\circ - A) = 1$.
3. If $3 \cot \theta = 2$, find the value of $\frac{4 \sin \theta - 3 \cos \theta}{2 \sin \theta + 6 \cos \theta}$.
4. If $\sin A = \cos A$, where $0 < A < 90^\circ$, find the value of A .
5. Evaluate $\frac{\tan 65^\circ}{\cot 25^\circ}$.
6. Given $15 \cot A = 8$, find the value of $\sin A$.
7. Find the value of $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$.
8. Prove that $\sec^2 \theta - 1 = \tan^2 \theta$.

Section C: Short Answer (3 marks each)

1. If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find the values of A and B .
2. Evaluate the following expression: $\frac{2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ}{\sin 30^\circ + \cos 60^\circ}$
3. If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, show that $\tan \theta = \frac{1}{\sqrt{3}}$.
4. If $\sqrt{3} \tan \theta = 1$, find the value of $\sin^2 \theta - \cos^2 \theta$.
5. Prove that $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A$.
6. If $\tan A = \frac{3}{4}$, find all other trigonometric ratios for angle A .

Section D: Long Answer (4-5 marks each)

1. Prove the following trigonometric identity: $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \csc \theta$.
2. If $\sec \theta + \tan \theta = p$, show that $\frac{p^2 - 1}{p^2 + 1} = \sin \theta$.
3. Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \csc \theta$.
4. If $\sec \theta + \tan \theta = p$, prove that $\sin \theta = \frac{p^2 - 1}{p^2 + 1}$.